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Mutagenic effect of heavy metal mercury (hg) on plant *Iberis amara* l.

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Abstract : Present investigation was carried out to find out the mitotic and meiotic chromosomal abnormalities induced in the M_1 and M_2 generation of two varieties Hyacinth flowered and Iceberg of *Iberis amara* L. by heavy metal Mercury (Hg) which is one of the most important environmental pollutant in today's world. A wide spectrum of mitotic and meiotic chromosomal aberrations like elongated cells, stickiness, laggards, bridges and binucleate cells etc. were encountered in both the varieties after the treatment with different concentrations of Mercury. The result concluded that treatment of *Iberis amara* L. with Mercury does not show a dose dependent increase or decrease in mitotic and meiotic chromosomal aberration in both the varieties. Lower treatment of heavy metal Mercury have influenced less chromosomal damage and would be suitable for inducing desirable mutations in *Iberis amara* L.

Keywords: *Iberis amara* L., Chromosomal abnormalities., Heavy metal., Stickiness., Laggards

INTRODUCTION

Industrialization has provided livelihood and opportunities to millions in urban India. However it has also enhanced the level of toxic heavy metals in the environment posing a potential health hazard for all living organism. [1, 2] Heavy metals are among foremost among pollutants and suspected carcinogens.[3, 4]

The present investigation was planned to study the mutagenic potentialities of heavy metals Hg as revealed by mitotic and meiotic abnormalities of *Iberis amara* L. plant which is a medicinal plant used in traditional medicine since ancient times.

MATERIALS AND METHOD

Seeds of two varieties of *Iberis amara* L. Hyacinth flowered and Iceberg were treated with five different concentrations (0.2, 0.4, 0.6, 0.8 and 1.0 ppm) of Mercury [II] chloride and young buds were collected from the plants grown after treatment. For mitotic studies ovaries separated from the buds were used and for meiotic studies pollen separated from buds were used. For staining 2% acetocarmine was used and slides were prepared by squash techniques. The cytological aberrations were studied and recorded.

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RESULT AND DISCUSSION

Treatment of both varieties of *Iberis amara* L. with different concentration of heavy metals induced high percentage of mitotic and meiotic abnormalities in both M_1 and M_2 generation.

Table 1 and 2 demonstrates the types and percentage of different mitotic abnormalities observed due to treatment with heavy metal Mercury(Hg) in M_1 and M_2 generation. Among the mitotic abnormalities noticed in both the varieties were elongated cells, ring chromosomes, stickiness, translocation chain, binucleate and micronuclei. A decrease in mitotic index was observed for M_1 and M_2 generation after the treatment which was not dose dependent. The reason for this type of result may be that the members of Brassicaceae family are actively hyper-accumulatory and competent to take heavy metals from the environment.[5]

Among the meiotic abnormalities noticed in present study as depicted in Table 3 and 4 were univalents, stickiness, unorientation, bridges, lagging chromosomes

and micronuclei.

Stickiness of the chromosomes was observed in both mitotic and meiotic study can be considered as most dominating abnormality. It has been suggested [6] that stickiness is a type of physical adhesion involving mainly proteinacious matrix of chromatin material.

Lagging chromosomes and bridges were also observed at a considerable percentage in different mitotic and meiotic stages (Table 1 - 4). The formation of bridges could be attributed to chromosomal stickiness [7] and to chromosome breakage and reunion. [8] Induction of lagging could be attributed to failure of normal organization and function of spindle apparatus. [6]

Unequal separation of chromosomes at anaphase of mitosis was observed frequently and such type of abnormalities is due to loss of microtubules of spindle fibres. [9]

Occurrence of ring chromosomes in this study after treatment with Hg in metaphase stage of mitosis may arise due to breaks in same chromosomes or due to telomeric losses due to which chromosome ends becomes sticky and fuse together. [10] Micronuclei observed during M_1 and M_2 generation in telophase stage of mitosis and meiosis in both varieties od *Iberis amara* L. may originates from lagging chromosomes at anapahase- telophase or from acentric fragments. [11] While multinucleate cells observed in telophase I of meiosis might be attributed to pairing due to translocation and inversion. [12]

It can be concluded from the present findings that mercury is a toxic agent in plant and causes significant increase in frequency of aberration.

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